Progressive Myoclonic Epilepsy Type 11 (EPM11): A Comprehensive Disease Characteristics Report

Disease: Progressive Myoclonic Epilepsy Type 11 (EPM11) **MONDO ID:** MONDO:0030034 · **OMIM:** #618876 · **Causal gene:** *SEMA6B* (HGNC:7357) **Category:** Mendelian (monogenic), autosomal dominant

Summary

Progressive Myoclonic Epilepsy Type 11 (EPM11) is an ultra-rare, autosomal-dominant progressive myoclonic epilepsy / developmental-and-epileptic-encephalopathy caused by heterozygous, usually *de novo*, **truncating variants clustered in the last exon (exon 17) of *SEMA6B*** (semaphorin-6B). The defining molecular feature is that these variants fall within the nonsense-mediated mRNA decay (NMD)-insensitive region of the transcript. Rather than being degraded, the mutant mRNA is translated into a **stable truncated protein lacking the intracellular domain**, which acts through a **toxic gain-of-function / dominant-negative mechanism**. This distinguishes EPM11 from simple haploinsufficiency disorders and explains why the *de novo* truncating variant class is so tightly disease-associated (PMID: 32169168; PMID: 36719163).

Mechanistically, SEMA6B is a transmembrane class-6 semaphorin that functions as an **axon-guidance receptor** binding floorplate-derived **PlexinA2** (and, in hippocampal mossy fibers, **PlexinA4**). The intracellular domain lost in disease truncations is essential for its receptor-signaling function. Disruption of SEMA6B signaling impairs axon guidance, dendritic-spine maturation, and synaptogenesis across cortical, hippocampal, and cerebellar circuits—including GABAergic interneurons—producing an excitation/inhibition imbalance that drives cortical hyperexcitability, seizures, and myoclonus, alongside developmental regression, intellectual disability, and ataxia (PMID: 25209245; PMID: 35604360; PMID: 20484647).

Clinically, EPM11 typically presents with developmental regression around age 2 after relatively normal early development, followed by polymorphic and often refractory seizures, progressive myoclonus, ataxia, spasticity, loss of independent ambulation, and severe cognitive impairment. The known worldwide caseload is tiny—approximately **36 reported patients**—spanning a phenotypic spectrum from classic PME to non-syndromic intellectual disability. There is **no disease-modifying or curative therapy**; management is symptomatic anti-seizure/anti-myoclonic polytherapy with supportive care, and prognosis is poor with lifelong, progressive disability (PMID: 39419291; PMID: 30999254).

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1. Disease Information

Overview. EPM11 is a monogenic progressive myoclonic epilepsy (PME) belonging to the group of developmental and epileptic encephalopathies. It is characterized by infantile/early-childhood onset, developmental regression, progressive myoclonus, polymorphic and frequently pharmacoresistant seizures, cerebellar and pyramidal/extrapyramidal signs, and severe cognitive impairment.

Key identifiers:

Resource	Identifier
OMIM	#618876 (Epilepsy, progressive myoclonic, 11)
MONDO	MONDO:0030034
Gene	*SEMA6B* (HGNC:7357); OMIM gene *608873
Inheritance	Autosomal dominant (ClinGen gene–disease validity: **Definitive**)
ICD-11	8A62 (Progressive myoclonic epilepsy) category
MeSH	Myoclonic Epilepsies, Progressive (D020191)

Synonyms / alternative names: EPM11; SEMA6B-related progressive myoclonic epilepsy; SEMA6B-related developmental and epileptic encephalopathy; progressive myoclonic epilepsy-11.

Source of information. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet, ClinGen) and from **individual patient case reports and small case series** in the primary literature, rather than from large EHR cohorts. Given the extreme rarity, the evidence base is dominated by case-level descriptions.

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2. Etiology

Primary cause — genetic. EPM11 is a purely monogenic disorder caused by heterozygous pathogenic variants in *SEMA6B*. The predominant and best-validated class is **truncating (frameshift/nonsense) variants clustered in the last exon (exon 17)**, which escape NMD and produce a toxic truncated protein (PMID: 32169168; PMID: 36719163).

Genetic risk factors. The causal variants are themselves the disease determinant. Most arise *de novo*; variants are absent from population databases such as gnomAD, consistent with pathogenicity. No susceptibility loci or modifier genes have been established.

Environmental risk factors. None implicated. No toxic, infectious, lifestyle, occupational, dietary, radiation, or age/sex-related environmental risk factors are known to contribute. EPM11 is not triggered by environmental exposure.

Protective factors. No genetic or environmental protective factors or modifier alleles are known. The only outcome-modifying factor identified is effective seizure control (see Prognosis).

Gene–environment interactions. None known; as a Mendelian disorder with high penetrance of the classic truncating variants, no GxE interactions have been characterized.

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3. Phenotypes

EPM11 is a multi-domain neurological disorder. Core phenotypes and suggested HPO terms:

Phenotype	HPO term (suggested)	Type	Onset	Progression	Frequency
Myoclonic seizures / myoclonus	HP:0001336 / HP:0001337	Clinical sign	Early childhood	Progressive, worsening	Core / near-universal
Progressive myoclonic epilepsy	HP:0007334	Clinical syndrome	~18 mo–6 yr	Progressive	50% of spectrum (18/36)
Developmental regression	HP:0002376	Sign	~2 yr	Progressive	Common
Intellectual disability (severe)	HP:0010864	Sign	Childhood	Progressive	Severe; often nonverbal
Polymorphic seizures (atonic, absence, focal, GTCS, negative myoclonus)	HP:0002197, HP:0002121, HP:0002384, HP:0002069	Clinical sign	18 mo–6 yr	Refractory in ~50%	Core
Ataxia / cerebellar signs	HP:0001251	Sign	1st–2nd decade	Progressive	Common
Spasticity / pyramidal signs	HP:0001257	Sign	1st–2nd decade	Progressive	Common
Tremor	HP:0001337	Sign	1st–2nd decade	Progressive	Common
Loss of ambulation	HP:0002540	Functional	1st–2nd decade	Progressive	Frequent
Non-convulsive status epilepticus	HP:0011146	Clinical sign	Variable	Episodic	Reported

Phenotype characteristics. Onset of developmental regression occurs around **2 years of age** after relatively normal early development; various seizure types emerge between 18 months and 6 years and are often refractory. Over the first and second decades, pyramidal,

extrapyramidal, and cerebellar signs accumulate. Cognitive impairment is **severe** (few words or non-verbal). The phenotype is **progressive** and does not correlate with variant type or location (PMID: 32169168; PMID: 36719163; PMID: 35604360).

Phenotypic spectrum. Cordovado et al. (2022) showed the spectrum extends beyond classic PME: *"until now, SEMA6B variants have mainly been reported in patients with progressive myoclonic epilepsy, our study indicates that the clinical spectrum is wider and also includes non-syndromic ID without epilepsy or myoclonus"* (PMID: 35604360). The 2025 systematic review (n=36) subdivided presentations into **PME (50%, 18/36), developmental and epileptic encephalopathy (36%, 13/36), neurodevelopmental disorder (11%, 4/36), and epilepsy (3%, 1/36)** (PMID: 39419291).

Quality of life impact. Severe. Progressive loss of ambulation, non-verbal status, refractory seizures, and myoclonus produce profound dependence in daily functioning and lifelong disability. EPM11 can clinically mimic Angelman syndrome (PMID: 42125992).

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4. Genetic / Molecular Information

Causal gene. *SEMA6B* (semaphorin-6B), HGNC:7357, located on chromosome 15q21.1, encoding a transmembrane class-6 semaphorin. UniProt: Q9H3T3.

Pathogenic variants. - **Predominant class:** Truncating (frameshift and nonsense) variants clustered in the **last exon (exon 17)**, which fall in the NMD-insensitive region. These escape nonsense-mediated decay and produce a stable protein lacking the intracellular domain (PMID: 32169168; PMID: 36719163). - **Rarer classes:** Missense variants (e.g., c.432C>G p.Cys144Trp, reported in an adult-onset case) and non-last-exon splice-site variants (e.g., **c.1680-2A>G**, NM_032108.4) producing diverse aberrant transcripts with **incomplete penetrance** (PMID: 42696975).

Variant classification (ACMG/AMP). Recurrent *de novo* exon-17 truncating variants are classified **pathogenic/likely pathogenic**, supported by genome-wide statistical enrichment in developmental and epileptic encephalopathy trios and absence from population databases.

Allele frequency. The causal *de novo* variants are **absent from gnomAD** and other population databases, consistent with pathogenicity.

Somatic vs germline. Variants are **germline** (constitutional), usually *de novo*; rare inherited (autosomal-dominant transmitted) cases exist.

Functional consequences. Gain of function / dominant-negative. Castellotti et al. state: *"The exon 17 is mainly involved, with truncating mutations causing the production of aberrant proteins with toxic gain of function"* (PMID: 36719163). The truncated protein lacks the intracellular domain required for receptor signaling and mislocalizes subcellularly, reducing dendritic spine maturation (PMID: 35604360).

Modifier genes / epigenetics / chromosomal abnormalities. None established. No DNA methylation or histone-modification changes have been characterized for EPM11, and no large-scale chromosomal rearrangements are implicated (though a 12p13.32 deletion appears in the broader PMA differential, unrelated to *SEMA6B*).

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5. Environmental Information

Not applicable. EPM11 is a monogenic disorder with no identified environmental, lifestyle, toxic, occupational, radiation, or infectious contributing factors. No infectious agents trigger or cause the disease.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **de novo (usually) truncating variant in the last exon (exon 17) of *SEMA6B*** arises → leads to a premature stop codon in the NMD-insensitive region.
2. Because the variant lies in the NMD(-) region, the mutant mRNA **escapes nonsense-mediated decay** → results in a **stable truncated SEMA6B protein lacking the intracellular domain** (*inferred from NMD rules + observed protein; demonstrated in overexpression systems*).
3. The truncated protein exerts a **dominant-negative / toxic gain-of-function effect** on SEMA6B signaling (and mislocalizes subcellularly) → results in disrupted SEMA6B–PlexinA2 (and PlexinA4) receptor signaling.
4. Disrupted semaphorin–plexin signaling → leads to **impaired axon guidance, reduced dendritic-spine density/maturation, and defective synaptogenesis**.

5. Faulty wiring across **cortical, hippocampal (mossy fiber), and cerebellar circuits, including GABAergic interneurons** → results in abnormal neural connectivity. **The mechanism branches here:**
6. **Branch 6a (epileptogenesis):** reduced inhibition / excitation–inhibition imbalance → **cortical hyperexcitability** → myoclonus, polymorphic and refractory seizures.
7. **Branch 6b (neurodevelopmental):** abnormal connectivity and ongoing circuit dysfunction → **developmental regression, severe intellectual disability, ataxia, and progressive pyramidal/extrapyramidal signs**.
8. Progressive accumulation of circuit dysfunction over the first two decades → leads to loss of ambulation, non-verbal status, and lifelong disability.

(Steps 1–3 and the gain-of-function nature are directly supported; the GABAergic/E-I-imbalance link in branch 6a is partly inferred from expression data and general PME pathophysiology.)

Supporting detail

- **Molecular pathways:** Semaphorin–plexin axon-guidance signaling. SEMA6B binds **PlexinA2** at the midline and **PlexinA4** in the hippocampus. GO:0007411 (axon guidance); GO:0071526 (semaphorin-plexin signaling pathway). *"we demonstrate that Sema6B binds to floorplate-derived plexin A2 (PlxnA2) for navigation at the midline"* (PMID: 25209245).
- **Protein dysfunction:** Loss of the intracellular domain abolishes receptor signaling: *"Truncated Sema6B lacking the intracellular domain was unable to rescue the loss-of-function phenotype, confirming a receptor function of Sema6B"* (PMID: 25209245). Patient variants mislocalize and reduce spine density: *"The overexpression of some variants leads to a subcellular mislocalization of SEMA6B protein in HEK293T cells and to a reduced spine density owing to loss of mature spines in neuronal cultures"* (PMID: 35604360).
- **Cellular processes:** Dendritic spine formation/maturation (GO:0060996, GO:0060997), synaptogenesis, neuronal migration/wiring. Cell types: cortical excitatory neurons (CL:0000679), GABAergic inhibitory interneurons (CL:0000617), cerebellar Purkinje cells (CL:0000121), hippocampal pyramidal neurons.
- **Subcellular compartments:** Plasma membrane (transmembrane receptor), dendritic spine, axonal growth cone; disease-associated mislocalization.
- **Metabolic / immune involvement:** None characterized; EPM11 is not a metabolic or immune-mediated PME.
- **Omics profiling:** No disease-specific transcriptomic, proteomic, metabolomic, or lipidomic signatures have been published; mechanistic data derive from targeted functional assays.

7. Anatomical Structures Affected

Organ / system level. Primary involvement is the **central nervous system** (UBERON:0001017). Key regions: **cerebral cortex** (UBERON:0000956), **cerebellum** (UBERON:0002037, Purkinje cells), **hippocampus** (UBERON:0002421, mossy fiber projections). Body system: **nervous system** (UBERON:0001016). No primary non-neurological organ involvement; secondary effects (e.g., immobility-related complications) follow from progressive disability.

Tissue / cell level. Nervous tissue. Affected cell populations: cortical excitatory neurons (CL:0000679), GABAergic interneurons (CL:0000617), cerebellar Purkinje cells (CL:0000121), hippocampal neurons. SEMA6B is highly expressed across these neuronal populations.

Subcellular level. Plasma membrane / transmembrane receptor localization; dendritic spines; axonal growth cones. GO cellular components: GO:0005886 (plasma membrane), GO:0043197 (dendritic spine).

Localization / lateralization. Diffuse, **bilateral** CNS involvement. EEG shows focal, multifocal, and diffuse abnormalities (see Diagnostics); brain MRI is often normal or non-specific.

8. Temporal Development

- **Onset:** Developmental regression around **age 2** after relatively normal early development; **insidious/subacute** onset. Seizures emerge between **18 months and 6 years**. A rare adult-onset missense case exists.
- **Progression:** Chronic and **progressive**. Over the first and second decades, pyramidal, extrapyramidal, and cerebellar signs worsen (spasticity, loss of ambulation, myoclonus, tremor, ataxia). Seizures frequently refractory (~50% pharmacoresistant). Course is lifelong.
- **Stages:** Early (regression + seizure onset) → intermediate (progressive myoclonus, motor decline) → advanced (loss of ambulation, non-verbal, severe disability).
- **Patterns:** No spontaneous remission; seizures episodic but underlying neurodegeneration is relentlessly progressive. The **critical window** for intervention is early seizure control, which is associated with improved neurodevelopmental outcomes (PMID: 42696975).

9. Inheritance and Population

- **Epidemiology:** Ultra-rare. Only **~36 patients reported worldwide** (2025 systematic review); no prevalence/incidence estimates exist (PMID: 39419291).
- **Inheritance: Autosomal dominant**, mostly **de novo**. ClinGen gene–disease validity **Definitive**. Rare inherited (AD-transmitted) families exist.
- **Penetrance:** High/complete for classic exon-17 truncating variants; **incomplete penetrance** documented for a splice-site variant family (c.1680-2A>G) — *"a novel heterozygous donor splice-site variant NM_032108.4 (SEMA6B): c.1680-2A>G segregating with autosomal dominant EPM1 in a family exhibiting marked incomplete penetrance"* (PMID: 42696975).
- **Expressivity:** Variable (PME to non-syndromic ID), not correlated with variant type/location.
- **Anticipation / mosaicism / founder effects / consanguinity:** No genetic anticipation (not a repeat-expansion disorder); germline mosaicism theoretically possible but not systematically documented; no founder effect or geographic clustering. The original 4 patients were ethnically diverse (2 Japanese, 1 Israeli, 1 Malaysian).
- **Carrier frequency:** Not applicable for a mostly *de novo*, dominant disorder.
- **Sex ratio / age distribution:** No sex predilection reported; affected individuals are predominantly children and young adults.

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10. Diagnostics

Diagnostic approach — genetic. Diagnosis relies on **molecular genetic testing**: exome (WES) or genome sequencing (WGS), or a multigene **epilepsy/PME panel** detecting the heterozygous *SEMA6B* truncating variant (typically exon 17, usually *de novo*). There is **no specific biochemical biomarker**, and brain MRI is often normal or non-specific (PMID: 32169168).

Test	Utility in EPM11
WES / WGS	High — primary diagnostic modality
Epilepsy / PME multigene panel	High — *SEMA6B* now included on PME panels
Single-gene *SEMA6B* sequencing	Useful if phenotype strongly suggestive
EEG	Supportive — focal/multifocal spikes, diffuse irregular spike-wave, high-amplitude slow waves
Brain MRI	Often normal/non-specific; used to exclude other causes
Biochemical/metabolic biomarkers	Not informative (no metabolic signature)
CMA/karyotype/FISH	Low yield (not a copy-number/structural disorder)
mtDNA / repeat-expansion testing	Used to exclude PME mimics (e.g., MERRF, EPM1), not positive in EPM11

Clinical criteria & differential diagnosis. Diagnosis integrates the PME clinical syndrome with a confirmatory *SEMA6B* variant. Differential diagnosis includes other PMEs (Unverricht-Lundborg disease/EPM1, Lafora disease, MERRF, neuronal ceroid lipofuscinoses, sialidosis, North Sea PME/GOSR2) and **Angelman syndrome**, which EPM11 can mimic (PMID: 42125992). Progressive myoclonus ataxia (PMA) overlaps on a PMA-PME spectrum (PMID: 38844245).

Screening. No population newborn or carrier screening exists (ultra-rare, mostly *de novo*). Cascade testing is only relevant in the rare inherited families.

11. Outcome / Prognosis

Prognosis is poor. EPM11 is a progressive neurodegenerative disorder (OMIM #618876). Over the first and second decades, affected individuals develop worsening pyramidal, extrapyramidal, and cerebellar signs—spasticity, **loss of independent ambulation**,

myoclonus, tremor, ataxia—and severe cognitive impairment, becoming **non-verbal** (PMID: 32169168). Seizures are frequently **refractory/pharmacoresistant** (~half of 36 patients) (PMID: 39419291).

- **Survival/mortality:** Specific survival and mortality statistics are **not established** given extreme rarity.
- **Morbidity/disability:** Severe, progressive, lifelong. Quality of life severely impaired.
- **Recovery potential:** None; the disorder does not remit and disability accumulates.
- **Prognostic factors:** The principal **modifiable** factor is seizure control — *"effective anti-seizure medication was associated with improved neurodevelopmental outcomes"* (PMID: 42696975).
- **Prognostic biomarkers:** None validated. Cortical hyperexcitability (giant SEPs / EMG-EEG correlate of myoclonus) may have prognostic value within the broader PMA-PME spectrum (PMID: 38844245).

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12. Treatment

No disease-modifying or curative therapy exists. Management is **symptomatic**, centered on anti-seizure medications (ASMs), as for progressive myoclonus epilepsies generally (PMID: 32169168).

Pharmacotherapy (NCIT terms suggested):

Drug	Class	Role in PME/EPM11	NCIT
Valproic acid	Broad-spectrum ASM	Mainstay for GTCS and myoclonus	NCIT:C935
Clonazepam	Benzodiazepine	Mainstay anti-myoclonic	NCIT:C577
Levetiracetam	SV2A modulator	Effective add-on for myoclonus	NCIT:C29102
Topiramate	Broad-spectrum ASM	Add-on option	NCIT:C921

Lasek-Bal et al. describe the standard polytherapy strategy in PME: *"ULD patients require anti-epileptic polytherapy, mostly benefitting from managing GTCS and myoclonus with valproic acid and clonazepam treatment. Patients may benefit from add-on therapy with levetiracetam or topiramate"* (PMID: 30999254).

Drugs to avoid. Sodium-channel blockers (phenytoin, carbamazepine, lamotrigine, vigabatrin) can **worsen myoclonus** and are generally avoided in PME.

Emerging / cross-PME insights. In related PME models, GABA-ergic agents (sodium barbital, clonazepam, and the neurosteroid **ganaxolone**) suppressed seizures more effectively than valproate/levetiracetam in a GOSR2 North Sea PME *Drosophila* model, highlighting a potential GABA-targeted direction (PMID: 38781737). Combination levetiracetam + clonazepam is effective for myoclonus in MERRF (PMID: 30334528). These are extrapolations, not EPM11-specific trials.

Advanced/experimental therapeutics. No gene therapy, RNA-based therapy, cell therapy, or targeted molecular therapy exists or is in trials for EPM11. The toxic gain-of-function mechanism in principle makes **allele-selective knockdown (ASO/siRNA)** a theoretically rational—but currently unexplored—strategy.

Supportive/rehabilitative care. Physical, occupational, and speech therapy; management of spasticity, nutrition, and mobility; multidisciplinary neurological care.

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13. Prevention

- **Primary prevention:** Not possible for *de novo* disease. In rare inherited families, **genetic counseling** and reproductive options (preimplantation genetic testing, prenatal diagnosis) apply.
- **Secondary prevention:** Early molecular diagnosis and early effective seizure control, which may improve neurodevelopmental outcomes (PMID: 42696975).
- **Tertiary prevention:** Avoiding myoclonus-aggravating ASMs; managing complications of progressive disability.
- **Counseling:** Genetic counseling is central—recurrence risk is low for confirmed *de novo* cases (with residual germline mosaicism risk) but up to 50% in autosomal-dominant transmitted families with variable/incomplete penetrance.
- **Immunization / public health / environmental interventions:** Not applicable.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *SEMA6B* is conserved across vertebrates. Mouse *Sema6b* (NCBI Gene) and chick *Sema6B* orthologs have been functionally characterized.
- **Natural disease:** No naturally occurring EPM11-equivalent disease has been described in companion animals or wildlife (no OMIA entry for a *SEMA6B* epilepsy phenotype).
- **Comparative biology:** The axon-guidance function of SEMA6B is evolutionarily conserved —defined in **mouse** (hippocampal mossy fibers, PlexinA4-dependent) and **chick** (commissural axons, PlexinA2-dependent) (PMID: 20484647; PMID: 25209245).
- **Transmission / zoonotic potential:** Not applicable (genetic, non-communicable).

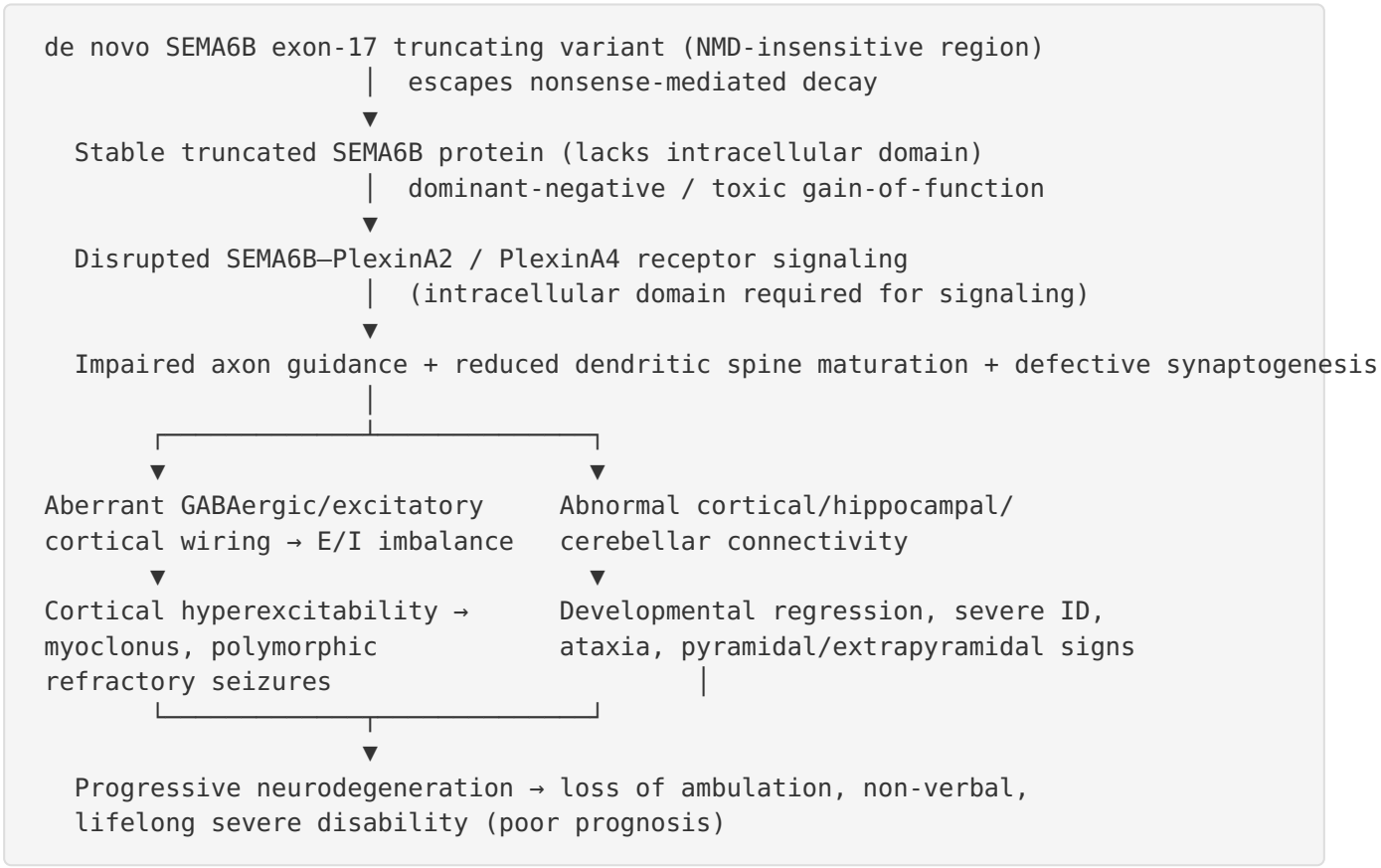
15. Model Organisms

Model	Type	Key finding	Relevance / limitation
Sema6b-null mouse	Mammalian knockout	Viable; aberrant hippocampal mossy fiber projections into stratum radiatum/oriens; additive with *Sema6a*	Models **loss-of-function**, not the human **gain-of-function** mechanism (PMID: 20484647)
Chick commissural neurons	In vivo RNAi/ rescue	SEMA6B acts as PlexinA2-binding axon-guidance receptor; intracellular domain essential	Established receptor role; not a disease model (PMID: 25209245)
HEK293T overexpression	In vitro	Patient variants mislocalize subcellularly	Variant functional assay (PMID: 35604360)
Primary neuronal culture (shRNA)	In vitro	Reduced dendritic spine density/maturation	Links SEMA6B to synaptic pathology (PMID: 35604360)
Chick embryo rescue assay	In vivo	Patient variants fail to rescue commissural axon pathfinding	Demonstrates variant pathogenicity (PMID: 35604360)

Tawarayama et al.: *"In Sema6B-deficient mice several mossy fibers aberrantly project to the stratum radiatum and the stratum oriens. The number of aberrant mossy fibers is increased in Sema6A;Sema6B double knock-out mice, indicating that Sema6A and Sema6B function additively to regulate proper projection of mossy fibers"* (PMID: 20484647).

Key model limitation: Existing knockouts recapitulate the **loss-of-function** axon-guidance phenotype but **do not model the dominant-negative/gain-of-function truncated protein** that drives human EPM11. A **knock-in mouse carrying a human exon-17 truncating variant** is the most important missing model.

Mechanistic Model / Interpretation



The model's **hinge** is step 3: the loss of the intracellular domain abolishes SEMA6B receptor signaling (PMID: 25209245), while NMD escape ensures the toxic protein is actually produced (PMID: 32169168). Upstream events (variant → NMD escape → truncated protein) are genetic

and well-demonstrated; downstream events (circuit mis-wiring → E/I imbalance → seizures) are supported by cellular/animal data but the specific GABAergic contribution to human epileptogenesis remains partly inferred.

Evidence Base

PMID	Title (abridged)	Evidence type	Contribution
32169168	*De Novo Truncating Variants in the Last Exon of SEMA6B Cause Progressive Myoclonic Epilepsy*	Human genetics (trio cohort)	Defines disease; NMD escape; dominant-negative/ GoF; original 4 patients
36719163	*Progressive myoclonus epilepsies due to SEMA6B mutations*	Human clinical/genetic	Exon-17 clustering; toxic GoF; core phenotype
35604360	*SEMA6B variants cause intellectual disability and alter dendritic spine density and axon guidance*	Human + in vitro/in vivo functional	Broad spectrum; spine density; mislocalization; chick rescue
25209245	*Semaphorin 6B acts as a receptor in post-crossing commissural axon guidance*	Model (chick)	Receptor role; PlexinA2 binding; intracellular domain essential
20484647	*Roles of semaphorin-6B and plexin-A2 in hippocampal mossy fibers*	Model (mouse)	KO phenotype; PlexinA4-dependent repulsion
39419291	*Systematic Review of SEMA6B-related Disorders*	Human systematic review (n=36)	Caseload, phenotypic subgroups, pharmacoresistance
42696975	*Novel SEMA6B splice-site variant (c.1680-2A>G)*	Human genetics/family	AD transmission; incomplete penetrance; ASM-outcome link
42125992	*SEMA6B-Related PME Mimicking Angelman Syndrome*	Human case	Differential diagnosis
30999254	*Unverricht-Lundborg disease: seizure management*	Human clinical	PME ASM polytherapy strategy
38844245	*Natural history of progressive myoclonus ataxia*	Human cohort	PMA-PME spectrum; cortical hyperexcitability prognostic value
38781737	*ASMs and GABA in North Sea PME*	Model (*Drosophila*)	GABA-targeted therapy direction (cross-PME)
30334528	*Levetiracetam+clonazepam in MERRF*	Human clinical	Combination anti-myoclonic efficacy (cross-PME)

Convergence vs. challenge. The genetic and functional lines of evidence converge strongly on the NMD-escape/gain-of-function model. The main internal tension is that the available **animal models are loss-of-function knockouts**, which demonstrate the gene's normal axon-guidance role but do not directly prove the *toxic gain-of-function* human mechanism—this is inferred from NMD biology, protein-domain rescue failure, and variant overexpression data.

Limitations and Knowledge Gaps

1. **Tiny evidence base (~36 patients).** No epidemiological estimates (prevalence/incidence), no survival/mortality statistics, and limited natural-history data.
2. **Mechanism inferred, not fully demonstrated in vivo.** No animal model expresses the human truncated gain-of-function protein; the dominant-negative mechanism is supported by overexpression and domain-deletion rescue data, not by an endogenous knock-in.
3. **GABAergic/E-I imbalance link is partly inferred.** The specific contribution of SEMA6B dysfunction in GABAergic interneurons to human epileptogenesis has not been directly measured in patient tissue.
4. **No omics signatures.** No disease-specific transcriptomic, proteomic, or metabolomic profiles exist for EPM11.
5. **No disease-specific therapy or trial.** All treatment data are extrapolated from PME broadly; no NCT-registered EPM11 trial exists.
6. **Genotype–phenotype correlation is weak/absent**, complicating prognostic counseling, and incomplete penetrance in non-last-exon variants adds uncertainty.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse (or human iPSC-neuron/organoid) carrying a recurrent exon-17 truncating *SEMA6B* variant** to directly test the dominant-negative/gain-of-function model and establish a translational model that recapitulates the human disease (not just loss of function).
2. **Characterize E/I balance and GABAergic interneuron wiring** in patient-derived iPSC cortical organoids with electrophysiology/MEA to validate the cortical-hyperexcitability branch.

3. **Test allele-selective knockdown (ASO/siRNA) of the mutant truncated transcript** as a mechanistically rational, disease-modifying strategy for a gain-of-function disorder.
4. **Screen GABA-targeted agents (e.g., ganaxolone, clonazepam) vs. standard ASMs** in an EPM11-specific model, building on cross-PME findings (PMID: 38781737).
5. **Establish an international patient registry** to capture natural history, seizure-control-vs-outcome relationships, survival, and genotype–phenotype data.
6. **Systematic cortical-hyperexcitability phenotyping** (giant SEPs, EMG-EEG) across the SEMA6B PMA-PME spectrum to develop prognostic biomarkers.

Report compiled from 11 confirmed findings across 5 investigation iterations and 14 reviewed primary papers. Evidence source types are indicated throughout: human clinical/genetic, model organism (mouse, chick, Drosophila), and in vitro functional.
